

Software Requirements Specification

Version 3.0

Annotated Version

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Road Repair and Tracking System

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1.Introduction

1.1. Purpose

The Road Repair and Tracking System (RRTS) is designed to automate the process of managing road repair activities for the Public Works Department of a large city. The system aims to streamline the process of raising repair requests, prioritizing them based on severity and locality type, and allocating resources efficiently.

It facilitates scheduling of repair tasks by considering the availability of raw materials, machinery, and personnel. The system also enables administrators to update manpower and machine availability, ensuring dynamic scheduling adjustments.

Additionally, it provides statistical reports on completed and pending repairs, workforce utilization, and overall resource efficiency, assisting city officials in decision-making and urban infrastructure management.

1.2. Scope of Project

The scope of the RRTS includes:

- **Request Management:** Recording and managing repair requests raised by residents for various roads across the city.
- **Prioritization:** Assessing the severity of road conditions and assigning repair priority based on locality type (e.g., commercial, residential, busy areas).
- **Resource Management:** Estimating and managing raw material requirements, machinery needs, and personnel allocation for each repair task.
- **Scheduling:** Automatically scheduling repair tasks based on priority and availability of resources.
- **Reporting:** Generating statistical reports on repair activities, outstanding repairs, and utilization of manpower and machinery over time.
- **Administrative Control:** Allowing city administrators to update available resources (manpower and machinery) and triggering rescheduling when necessary.

1.3. Definition, Acronyms and Abbreviations

1.3.1 Definitions

- **Road Repair and Tracking System (RRTS):** A software solution designed to automate the road repair request management, resource allocation, repair scheduling, and reporting for the Public Works Department of a large city's Corporation.
- **Public Works Department (PWD):** A municipal department responsible for the construction, repair, and maintenance of public infrastructure such as roads, bridges, and public buildings.
- **Supervisor:** A designated staff member who assesses the severity of the road condition, determines repair priorities, estimates resource requirements (e.g., manpower, raw materials, and machines), and supervises the repair work.
- **City Corporation Administrator:** An administrator responsible for maintaining and updating the data regarding available manpower, machinery, and raw materials. They also modify resource availability when required.
- **Audit Trail:** A record of all actions and changes made to the system, such as resource allocation, repairs, and scheduling, to ensure transparency and accountability.

1.3.2. Acronyms

- **RRTS:** Road Repair and Tracking System
- **PWD:** Public Works Department
- **UI:** User Interface
- **DB:** Database
- **API:** Application Programming Interface
- **GUI:** Graphical User Interface
- **URL:** Uniform Resource Locator
- **SLA:** Service Level Agreement
- **ROI:** Return on Investment

1.3.3 Abbreviations

- **Req:** Request
- **Rep:** Repair
- **SLA:** Service Level Agreement
- **PR:** Priority
- **R&M:** Raw Material
- **Man:** Manpower
- **Mach:** Machinery
- **Est:** Estimated
- **Sup:** Supervisor
- **Admin:** Administrator
- **Sched:** Schedule
- **Stats:** Statistics

1.4 Glossary of Term

- **Repair Severity:** A measure of how badly a road is damaged. Typically categorized into low, medium, or high severity, influencing the priority and resources allocated to repairs.
- **Ticket:** A logged road repair request.
- **Machine/Equipment:** Machines or tools (e.g., asphalt mixers, road rollers, etc.) used in road repairs. These can be tracked in the system for availability and scheduling.
- **Manpower Requirements:** The number and type of personnel needed to carry out road repairs. This includes different roles such as laborers, machine operators, and supervisors.
- **Schedule Report:** A document generated by the system that lists all the scheduled repair tasks, the assigned resources, and the expected start and end dates for repairs.

- **Resource Availability:** The data entered by the city corporation administrator regarding the availability of raw materials, machinery, and manpower. Changes in resource availability can affect the scheduling of repairs.
- **Priority Level:** A classification that assigns importance to repair requests based on factors like urgency, location, and traffic load. Priority levels influence the order in which repair tasks are performed.

1.5 System Architecture

The system is designed to automate and streamline road repair activities. Key components of the system include:

- **Frontend/UI:** The graphical user interface through which clerks, supervisors, administrators, and the mayor interact with the system.
- **Backend/Processing:** The engine that processes data, such as scheduling repairs and resource allocation.
- **Database:** A centralized database that stores all relevant data such as repair requests, resource availability, repair schedules, and user roles.
- **Reporting Module:** Generates and outputs reports on the status of repairs, statistics on resources, and completed tasks.

1.6. References

- **IEEE 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
- City Corporation Road Repair Guidelines, 2023.
- **Software Engineering:** A Practitioner's Approach by Roger S. Pressman.
- **City Infrastructure Management Guidelines** – Government of India, Ministry of Urban Development, 2022.

1.7. Overview of Document

RRTS is a centralized software solution that digitizes the road repair workflow of a city corporation. The system allows residents to report road damages, which are then evaluated by supervisors for severity and priority.

Based on this assessment, the system schedules repair tasks while ensuring optimal use of resources. The city administrator can update resource availability, triggering automatic rescheduling when required.

The system generates comprehensive reports on road maintenance activities, assisting the mayor and city officials in monitoring progress and resource utilization. By automating these processes, RRTS improves efficiency, reduces response times, and enhances overall urban infrastructure management.

2.0 Overall Description

2.1 Product Perspective

The **Road Repair and Tracking System (RRTS)** automates the road repair process for the Public Works Department. It manages repair requests, resources, scheduling, and reporting efficiently.

The system integrates with branch offices, providing a centralized platform for tracking repairs, monitoring resources, and generating statistics.

Developed as a web-based application, it ensures accessibility and real-time data updates. It assists supervisors and administrators in decision-making and daily operations.

Key Features of the RRTS include:

- **Repair Request Management:** Allows residents to raise repair requests for roads, which are recorded and tracked by the system.
- **Repair Prioritization:** Supervisors can assess the severity of the damage and prioritize repairs based on various factors such as locality, type of damage, and traffic conditions.
- **Resource Scheduling:** The system will generate schedules for repair tasks based on the availability of manpower, machines, and materials.
- **Resource Management:** Tracks the availability and utilization of manpower and machinery for road repairs.
- **Report Generation:** Generate repair statistics, outstanding repairs, and resource utilization reports for administrators and city officials.
- **Real-time Updates:** Reflects changes in available resources and schedules instantly.

2.2 Development Environment and Constraints

Development Environment:

- **Programming Languages:**
 - **Frontend:** HTML, CSS, JavaScript, React.js
 - **Backend:** Flask or Spring Boot or Next.js or Node.js or Express.js
 - **Database:** MongoDB for storing requests, repair details, and resource information
- **Server:** Apache Tomcat or Render or Netlify (for deployment)
- **Operating System:** The system will be designed to be cross-platform, compatible with both Windows and Linux servers.
- **Web Browser Support:** Google Chrome, Mozilla Firefox, and Microsoft Edge will be supported for the frontend user interface.

Constraints:

- **Resource Limitations:** Ensure accurate scheduling based on available resources (manpower, materials, machines).
- **Data Security:** Adhere to security standards for protecting sensitive data (repair requests, resource management, city statistics).
- **User Training:** Provide training for supervisors, clerks, and administrators to effectively use the system.

2.3 Assumptions

- **Infrastructure Availability:** Assumes stable hardware and network infrastructure in city corporation and branch offices.
- **Timely Input from Administrators:** Assumes administrators will regularly update real-time data on available manpower and machines.
- **User Adoption:** Assumes training and effective use by clerks, supervisors, and administrators for managing road repairs.
- **Regulatory Standards:** Assumes current regulations for road repairs and resource management will remain unchanged during development and initial deployment.
- **Manual Inspections:** System automates scheduling and tracking of repairs but requires manual inspections by supervisors for assessing repair priorities.

System Objective: Aims to efficiently manage road repair tasks, ensuring timely and prioritized repairs while optimizing available resources.

3.0 Functional Requirements

The **Road Repair and Tracking System (RRTS)** automates road maintenance for the Public Works Department, allowing citizens to submit repair requests for better engagement and faster resolution. It enhances transparency and efficiency in addressing infrastructure issues.

Supervisors can prioritize and schedule repairs based on severity and resource availability, ensuring optimal manpower and machinery use. Administrators can monitor resource allocation, track progress, and make informed decisions.

3.1 Inputs

The system will receive several types of input data to manage and process repair requests, resource allocation, and scheduling. The inputs are categorized as follows:

3.1.1 Repair Requests

- **Description:** The system allows the public to submit repair requests, including details about the damaged road such as its location, the type of damage, and any additional comments or information from the user.
- **Attributes:**
 - Resident Name
 - Address or Road Location
 - Type of Damage (e.g., pothole, crack, surface damage)
 - Severity of Damage (Low, Medium, High)
 - Date of Request

3.1.2 Road Condition Assessment

- **Description:** Supervisors will input their evaluation of the severity of the road condition based on their inspection.
- **Attributes:**
 - Road Location (linked to the request)
 - Severity Assessment (Low, Medium, High)
 - Additional Notes (e.g., type of surface, specific repair needs)
 - Estimated Raw Materials Requirement (types and quantity)
 - Estimated Machines and Equipment Required (types and quantity)
 - Estimated Personnel Requirements (number and types of workers)

3.1.3 Resource Availability

- **Description:** City administrators will input data on available resources . This data will be updated as needed.
- **Attributes:**
 - Available Manpower (number of workers by type: skilled, unskilled, etc.)
 - Available Machines (types and quantities)
 - Raw Materials (types and quantities, e.g., asphalt, cement)

3.1.4 Scheduling Data

- **Description:** Supervisors will provide input on the priority of each repair request, which will help determine the order of execution for repairs.
- **Attributes:**
 - Priority Level (High, Medium, Low)
 - Scheduled Start Date
 - Scheduled End Date

3.1.5 System Modifications by Administrators

- **Description:** Administrators may modify scheduling, resource availability, or priorities based on changing circumstances.
- **Attributes:**
 - Change Type (e.g., change in resource availability, change in priority)
 - Affected Repair Request or Resource
 - New Data (e.g., updated manpower, new priority)

3.2 Processing

The system will process the above inputs to generate schedules, track repair progress, and provide statistical reports. The processing steps will be as follows:

3.2.1 Request Entry and Verification

- **Description:** When a repair request is submitted by a user, the system will verify the data (e.g., location, type of damage) for completeness and accuracy.
- **Processing Steps:**
 1. A user submits a repair request.
 2. The system checks for any missing or invalid information.

3. If the request is complete and valid, it is saved to the database.
4. If any issues are found, an error message is displayed to the user.

3.2.2 Road Condition Assessment and Resource Estimation

- **Description:** The supervisor inputs the road condition assessment and estimates the required resources for the repair.
- **Processing Steps:**
 1. Supervisor enters severity and resource requirements.
 2. The system stores the severity and resource estimates.
 3. The system checks for available resources (manpower, machines, materials) and calculates the feasibility of the repair.

3.2.3 Priority Determination

- **Description:** Based on the road condition and the locality (commercial, busy, or deserted area), the supervisor determines the repair priority (High, Medium, Low).
- **Processing Steps:**
 1. The supervisor inputs the priority for the repair request.
 2. The system stores the priority level and calculates the optimal schedule based on priority.

3.2.4 Scheduling

- **Description:** The system schedules the repair tasks based on available resources, road conditions, and priority. The schedule also accounts for the availability of manpower, machines, and materials.
- **Processing Steps:**
 1. The system schedules repairs based on priority and resource availability.
 2. The system generates a schedule report for the supervisor, showing the planned start and end dates.
 3. If there are insufficient resources, the system will flag the repair as unassignable or suggest rescheduling.

3.2.5 Resource Tracking and Updates

- **Description:** The system tracks the real-time status of resources and updates the available manpower, machines, and raw materials as necessary.
- **Processing Steps:**
 1. City administrator enters updates to the resource database.
 2. The system re-calculates the schedule whenever resources are updated, ensuring that repairs are rescheduled if necessary.

3. The system flags any repairs that cannot proceed due to resource shortages.

3.2.6 Reporting and Statistics Generation

- **Description:** The system generates various reports for city officials, including the mayor and department administrators, on the status of repairs, outstanding tasks, and resource utilization.
- **Processing Steps:**
 1. The mayor or administrator requests specific reports (e.g., repairs completed in a given period, repairs pending, resource utilization).
 2. The system generates the requested report by querying the repair request and resource databases.
 3. The report is displayed to the officials or exported for further use (e.g., in CSV format).

3.3 Outputs

The system generates several outputs to facilitate decision-making, track progress, and provide transparency to stakeholders. These outputs include:

3.3.1 Repair Request Confirmation

- **Description:** After a repair request is entered into the system, a confirmation is provided to the resident and clerk.
- **Output:**
 - Confirmation Message (including repair request ID)
 - Estimated response time and next steps

3.3.2 Repair Schedule Report

- **Description:** A detailed report showing the schedule of repairs based on priority, resources, and availability.
- **Output:**
 - List of scheduled repairs (with start and end dates)
 - Resource assignments (manpower, machines, and materials)
 - Status of each repair task.

3.3.3 Resource Availability and Allocation Report

- **Description:** A report displaying the current resource availability and allocations across different repair tasks.
- **Output:**
 - Resource inventory (manpower, machines, and raw materials)
 - Resource allocation to specific repairs
 - Any conflicts or shortages in resource availability

3.3.4 Road Repair Status Report

- **Description:** A report detailing the status of all repair work, including completed, ongoing, and pending tasks.
- **Output:**
 - List of completed repairs
 - List of repairs in progress
 - List of pending repairs (with reasons for delay if applicable)

3.3.5 Statistical Reports

- **Description:** Various statistical reports for tracking road repairs and resource utilization.
- **Output:**
 - Total number of repairs carried out in a given period
 - Breakdown of repairs by type and severity
 - Utilization of manpower and machines over time
 - Outstanding repair work statistics

3.3.6 Alerts and Notifications

- **Description:** Notifications are sent to supervisors and administrators in case of schedule conflicts, resource shortages, or any updates that require attention.
- **Output:**
 - Alert messages (e.g., "Insufficient resources for repair ID: 123")
 - Notifications for rescheduled tasks.

4.Non-Functional Requirements

A non-functional requirement defines the qualities and constraints a system must meet to ensure performance, usability, security, scalability, and other critical attributes.

Unlike functional requirements, which specify what a system should do, non-functional requirements focus on **how** the system should perform.

4.1 *Performance Requirements*

4.1.1 Response Time

- The system must provide real-time updates for all user inputs, such as entering a repair request, resource availability updates, and repair scheduling.
- **Response Time for User Actions:**
 - For entering a repair request, the system should acknowledge and provide a confirmation within 3 seconds.
 - For scheduling a repair, the system should generate a schedule report within 5 seconds.
 - The system should process and display any resource updates (manpower, machines, or materials) in less than 5 seconds.
- **Reports Generation:**
 - The system should generate a statistical or repair status report within 30 seconds, even for large sets of data (e.g., over 500 repair requests).

4.1.2 Throughput

- The system should support multiple users accessing it concurrently, including residents, supervisors, and administrators.
- The system should handle a high volume of repair requests each month without any degradation in performance.

4.1.3 Scalability

- The system must be scalable to accommodate increasing numbers of repair requests, users, and resource updates as the city's population or size grows.

4.1.4 Availability

- The system should maintain high availability, ensuring continuous access every day, except during scheduled maintenance.

- In case of a failure or crash, the system should recover within 10 minutes with no data loss.

4.1.5 Reliability

- The system should ensure that it processes and stores all data accurately and without errors. Any inconsistencies should trigger an alert to the administrator or responsible personnel for investigation.
- The system must ensure data integrity for all repair requests, resource updates, and scheduling activities.

4.1.6 Load Balancing

- The system should have load balancing capabilities to handle spikes in traffic, ensuring that no single server is overwhelmed with requests, especially during peak usage times (e.g., when large repair updates or reports are generated).

4.2 User-friendliness

4.2.1 Interface Design

- The user interface (UI) of the system should be intuitive, easy to navigate, and accessible to users with various technical backgrounds.
- The system should employ consistent design patterns across all screens, ensuring familiarity and reducing the learning curve.
 - Navigation menus should be clearly labeled and easily accessible.
 - Important actions should be highlighted (e.g., “Create New Repair Request” or “View Scheduled Repairs”).

4.2.2 Ease of Use

- The system should minimize the number of clicks required for performing common actions (e.g., entering a repair request, updating resource data, generating a report).
- Forms for entering data (e.g., repair request or resource updates) should have clear and concise labels, with validation messages for incomplete or incorrect entries.

4.2.3 Accessibility

- The system should be accessible on both desktop and mobile platforms, ensuring that users can manage repairs and schedules while on the go.

- The system should support accessibility standards, including screen reader compatibility and color contrast for visually impaired users.

4.3 Security

4.3.1 Authentication and Authorization

- The system should implement role-based authentication to ensure that users can only access features and data relevant to their role (e.g., residents, supervisors, city administrators, mayor).
 - **Residents:** Can enter and update repair requests.
 - **Supervisors:** Can assess repair requests, determine priorities, allocate resources, and generate schedules.
 - **City Administrators:** Can modify resource availability and manage user accounts.
 - **Mayor/City Officials:** Can access reports and statistical data.
- All users must authenticate using a secure login mechanism (username and password). Passwords should be stored using a secure hashing algorithm (e.g., bcrypt or SHA-256).

4.3.2 Data Encryption

- Sensitive data (e.g., repair requests, user login credentials, resource data) should be encrypted during transmission (using HTTPS/SSL) and at rest (using database encryption).
- Any communication between the user's browser and the server should be protected by SSL/TLS protocols to prevent interception of sensitive data.

4.3.3 Role-based Access Control (RBAC)

- The system must implement RBAC to ensure that users can only access functionalities permitted for their role.
- For example:
 - **Residents** can input repair requests but cannot modify schedules or manage resources.
 - **Supervisors** can view and update repair priorities and schedules but cannot modify user accounts.
 - **City administrators** can modify resource availability but cannot modify repair priorities.

4.3.4 Logging and Audit Trails

- The system should maintain an audit trail of all user actions, including the creation of repair requests, updates to resources, changes in scheduling, and report generation.
- Logs should be stored securely and be accessible only to authorized personnel.

4.3.5 Backup and Data Recovery

- Regular backups of the system's database should be taken to prevent data loss in case of system failure.
- The backup should include all repair requests, scheduling data, and resource information.
- The system should provide the ability to restore data from backups with minimal downtime in case of a disaster or crash.

4.3.6 System Security Updates

- The system should allow for security patches and updates to be applied promptly without requiring significant downtime.
- Security vulnerabilities should be patched as soon as they are identified by the development or security team, with a process in place for testing and deploying fixes.

4.3.7 Protection from Denial of Service (DoS) Attacks

- The system should have measures in place to protect against DoS attacks or other attempts to overload the server, ensuring the system remains responsive even under heavy load.
- Rate limiting and other preventive measures should be in place to detect and mitigate unusual traffic patterns.

4.3.8 Session Management

- The system should enforce session timeouts after a period of inactivity to enhance security.

Session tokens should be securely stored and regenerated after each login to prevent session hijacking.

5.0 Data Dictionary

The Data Dictionary for the **Road Repair and Tracking Software (RRTS)** defines key data elements essential for automating road repair processes. It includes **Repair Request**, capturing details like Request ID, Road Name, Locality, Severity Level, and Priority.

The **Supervisor Report** stores Severity Assessment, Material Requirement, Machine and Manpower Estimates. The **Schedule Report** records Repair ID, Scheduled Date, Assigned Machines, Assigned Personnel, and is dynamically updated based on availability.

The **Resource Inventory** maintains Manpower, Machines, and Material Stock, which the **Administrator** updates as needed. Finally, **Statistical Reports** include Completed Repairs, Pending Repairs, Resource Utilization, enabling the **Mayor** to oversee repair efficiency.

5.1. Residents Table

Field Name	Data Type	Description
Resident_ID	INT	Unique identifier for each resident.
Name	VARCHAR	Name of the resident.
Address	VARCHAR	Address of the resident, including the suburb/area.
Contact_Number	VARCHAR	Contact phone number of the resident.

5.2. Supervisor Table

Field Name	Data Type	Description
Supervisor_ID	INT	Unique identifier for the supervisor.
Name	VARCHAR	Name of the supervisor.
Contact_Number	VARCHAR	Contact phone number of the supervisor.

Suburb	VARCHAR	Suburb where the supervisor works.
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5.3. City Corporation Administrator Table

Field Name	Data Type	Description
Admin_ID	INT	Unique identifier for the administrator.
Name	VARCHAR	Name of the administrator.
Contact_Number	VARCHAR	Contact phone number of the administrator.
City	VARCHAR	City governed by the City Corporation Administrator.

5.4. Mayor Table

Field Name	Data Type	Description
Mayor_ID	INT	Unique identifier for the mayor.
Name	VARCHAR	Name of the mayor.
Contact_Number	VARCHAR	Contact phone number of the mayor.
City	VARCHAR	City governed by the mayor.

5.5. Repair Request Management Table

Field Name	Data Type	Description
Request_ID	INT	Unique identifier for each repair request.
Resident_ID	INT	Unique identifier of the resident who raised the request.
Road_ID	INT	Unique identifier for the road requiring repair.
Request_Date	DATE	Date when the repair request was raised by the resident.

Priority	VARCHAR	Priority of the repair request (e.g., high, medium, low).
Type_of_Repair	VARCHAR	Type of repair needed (e.g., patching, resurfacing).
Requested_By	VARCHAR	Name of the resident or entity requesting the repair.

5.6. Road Condition Assessment Table

Field Name	Data Type	Description
Assessment_ID	INT	Unique identifier for each road condition assessment.
Request_ID	INT	The ID of the related repair request.
Supervisor_ID	INT	ID of the supervisor performing the assessment.
Severity	VARCHAR	Severity of the road condition (e.g., minor, moderate, major).
Assessment_Date	DATE	Date of the condition assessment.
Road_Type	VARCHAR	Type of the road (e.g., commercial, residential).
Damage_Type	VARCHAR	Type of damage (e.g., potholes, cracks, erosion).
Estimated_Repair_Time	INT	Estimated time to complete the repair (in days).

5.7. Resource Estimation Table

Field Name	Data Type	Description
Estimation_ID	INT	Unique identifier for each resource estimation.
Request_ID	INT	The related repair request ID.
Raw_Material_Type	VARCHAR	Type of raw material needed (e.g., asphalt, concrete).
Material_Quantity	DECIMAL	Quantity of raw material required for the repair (in kg, tons, etc.).
Machine_Type	VARCHAR	Type of machine needed (e.g., bulldozer, paver).

Machine_Quantity	INT	Number of machines required.
Personnel_Type	VARCHAR	Type of personnel needed (e.g., laborers, foreman).
Personnel_Quantity	INT	Number of personnel needed.

5.8. Repair Scheduling & Allocation Table

Field Name	Data Type	Description
Schedule_ID	INT	Unique identifier for each repair schedule.
Request_ID	INT	The related repair request ID.
Supervisor_ID	INT	The ID of the supervisor overseeing the repair work.
Scheduled_Start_Date	DATE	Scheduled date for the start of the repair work.
Scheduled_End_Date	DATE	Scheduled completion date of the repair work.
Actual_Start_Date	DATE	Actual start date of the repair work.
Actual_End_Date	DATE	Actual completion date of the repair work.
Repair_Status	VARCHAR	Current status of the repair (e.g., planned, in progress, completed).

5.9. Resource Management Table

Field Name	Data Type	Description
Resource_ID	INT	Unique identifier for each resource.
Resource_Type	VARCHAR	Type of resource (e.g., manpower, machine, raw material).
Available_Quantity	INT/DECIMAL	Current available quantity of the resource.
Used_Quantity	INT/DECIMAL	Quantity of resource used in the ongoing repairs.
Remaining_Quantity	INT/DECIMAL	Remaining quantity of the resource.

Resource_Status	VARCHAR	Status of the resource (e.g., available, under maintenance).
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5.10. Reporting & Analytics Table

Field Name	Data Type	Description
Report_ID	INT	Unique identifier for the report.
Report_Type	VARCHAR	Type of report (e.g., repair status, resource utilization).
Report_Date	DATE	Date when the report is generated.
Start_Date	DATE	Start date for the reporting period.
End_Date	DATE	End date for the reporting period.
Repair_Type	VARCHAR	Type of repair covered in the report (e.g., completed, pending).
Repair_Statistics	TEXT	Summary of statistics like total repairs, pending repairs, etc.
Resource_Utilization	TEXT	Statistics on resource utilization (e.g., manpower, machines).

6.0 Data Flow Diagram

The **Road Repair and Tracking System (RRTS)** is designed to automate the processes associated with road repairs in a city, ensuring efficient management and tracking of repair tasks. Here's a concise description of how the system functions across different components:

6.1. Level 0: Context Diagram

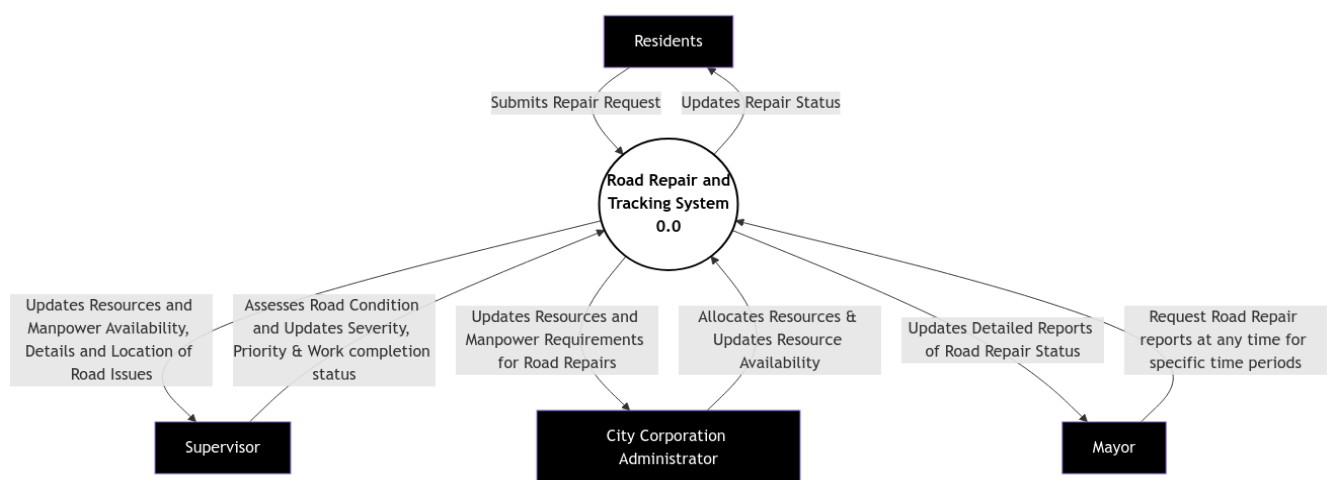
At the highest level, the system integrates external entities such as **Residents**, **Supervisors**, **City Corporation Administrators**, and the **Mayor**. Residents submit repair requests for road conditions, and the system processes these through the Supervisor, who assesses the severity and assigns priorities. The City Corporation Administrator provides resource availability, and the Mayor requests detailed reports for statistics and performance.

In the Level 0 DFD of the Road Repair and Tracking System (RRTS), the system is depicted as a single process, which receives inputs from various external entities, processes the data, and outputs relevant information. The main external entities involved are:

1. **Residents**: Residents of the city submit road repair requests through a simple user interface. They provide details such as the road location, nature of the damage, and urgency of the request. This input is sent to the system for processing.
2. **Clerk**: The clerk enters the road repair requests into the system. This is the initial data entry stage, where the requests are captured and stored for further processing by the supervisor.
3. **Supervisor**: The supervisor reviews each repair request and assesses the severity of the road condition. Based on this assessment, the supervisor determines the priority of the repair, estimates the materials, machines, and manpower required, and provides a work schedule to the system.
4. **Administrator**: The administrator inputs and maintains the data about the available manpower and machines. These values are continuously updated to ensure the scheduling of repairs is accurate and feasible. If there is a change in availability, the system needs to reschedule repairs accordingly.
5. **Mayor**: The mayor requests various statistics, such as the number of repairs completed, outstanding repairs, and the utilization of resources like manpower and machines. These statistics are provided by the system based on historical data and real-time tracking.

The central process in the Level 0 DFD is the **RRTS System**, which receives data from the residents, clerk, supervisor, and administrator, processes the requests, and outputs schedules, repair data, and statistics to the supervisor and mayor.

Level 0 Data Flow Diagram



6.2. Level 1: Data Flow Diagram

A Level 1 Data Flow Diagram (DFD) breaks down a high-level process into smaller sub-processes, showing the system's major functions, data flows, and external interactions. It provides a detailed view of how data is processed and flows between components within the system.

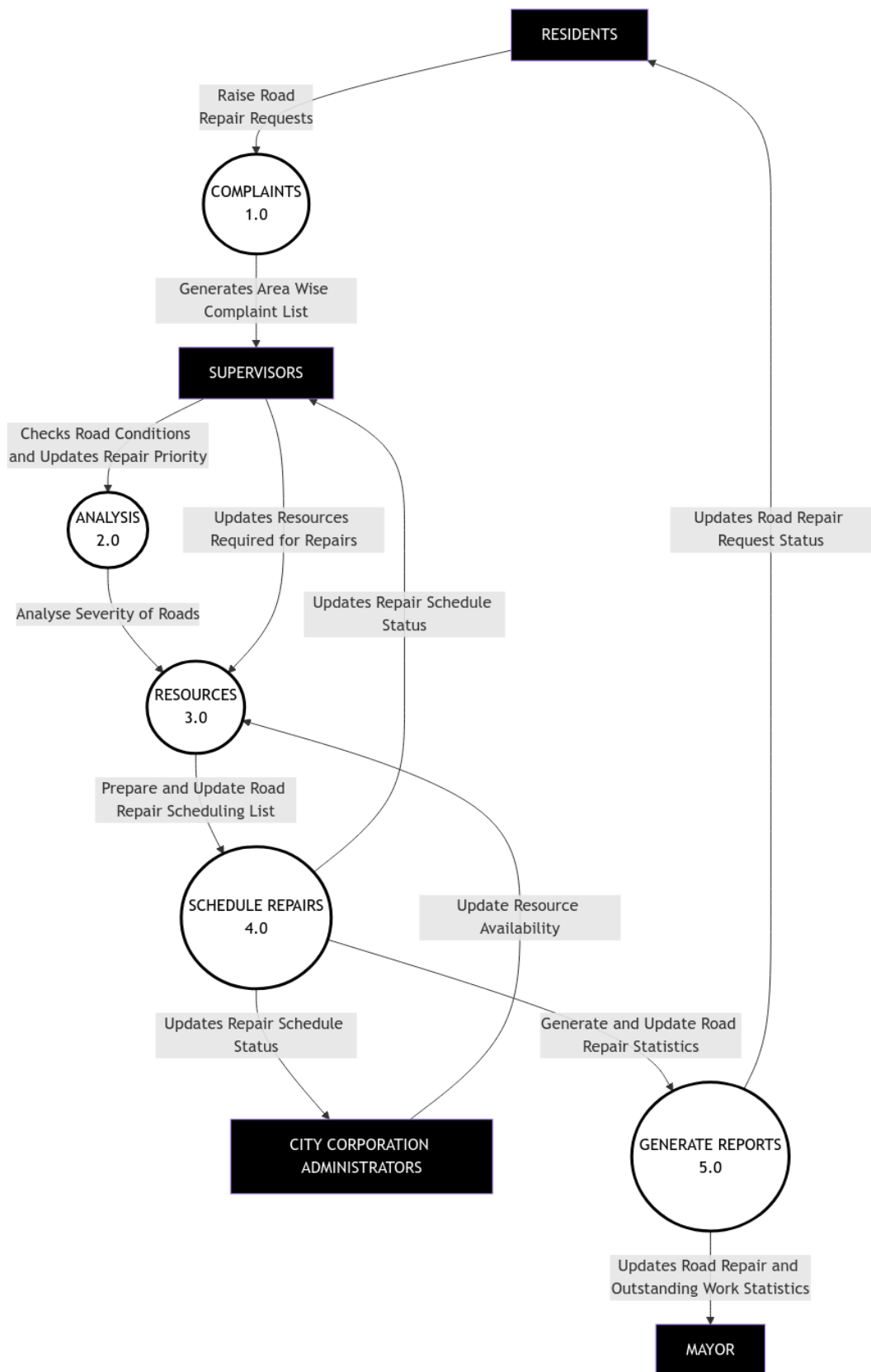
1. **Receive Repair Request:** Captures and stores repair requests from residents, including road condition and urgency.
2. **Assess Road Condition:** Supervisor assesses the severity of the road damage and assigns priorities for repair.
3. **Estimate Resources:** Based on the severity, the supervisor estimates the required manpower, machines, and materials.
4. **Schedule Repair Work:** Using available resources, the system schedules repairs based on priority and available resources.

5. **Update Resource Availability:** The administrator updates resource data, ensuring the scheduling reflects current availability.
6. **Generate Reports:** The system generates reports for the mayor, summarizing repairs, outstanding work, and resource utilization.

The Level 1 DFD of the Road Repair and Tracking System (RRTS) includes several key processes: capturing and validating road repair requests submitted by residents, assessing the road condition and determining repair priority by the supervisor, estimating resource requirements (materials, machines, manpower), scheduling repairs based on available resources, tracking resource availability and updates by the administrator, and generating reports and statistics for the mayor.

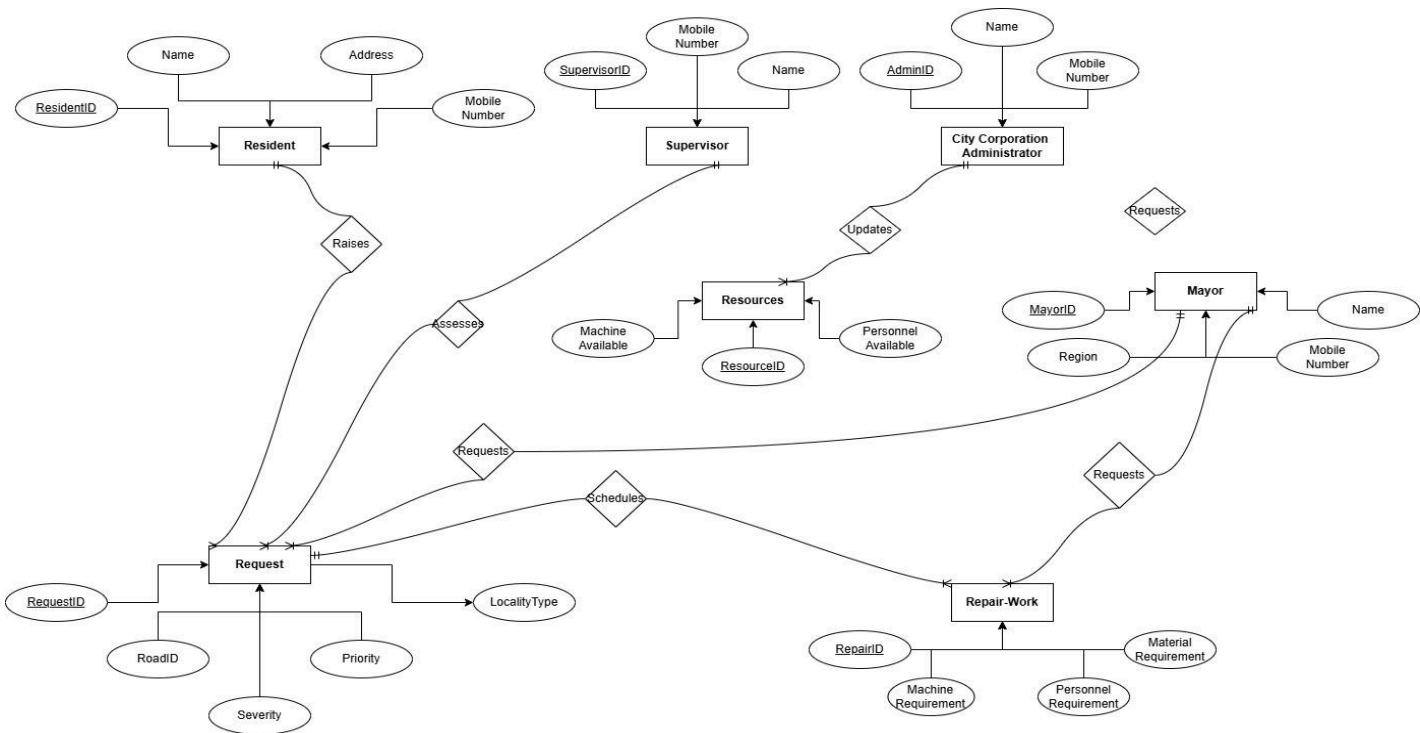
The system uses the collected data to manage repair tasks, allocate resources efficiently, and generate reports on repair progress and resource utilization. These processes ensure smooth operation, effective task management, and provide valuable insights for decision-making.

The **Level 1 DFD of RRTS** breaks down the system into processes that handle road repair requests, prioritize repairs, allocate resources, schedule tasks, and generate reports. This structured approach helps in managing repairs efficiently, ensuring resources are utilized optimally, and providing essential data for decision-makers like the supervisor and mayor.



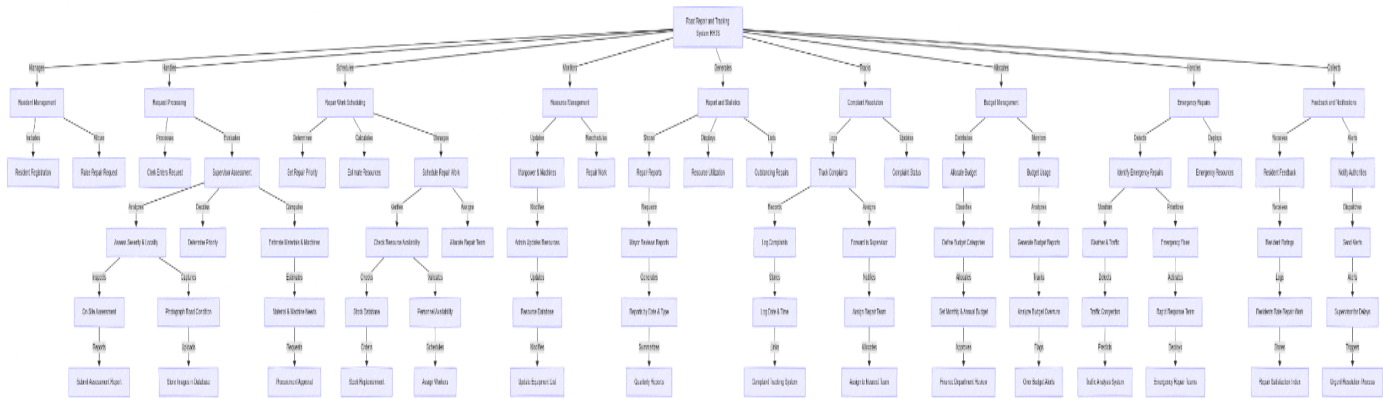
7.0 UML Diagrams

7.1 ER Diagram

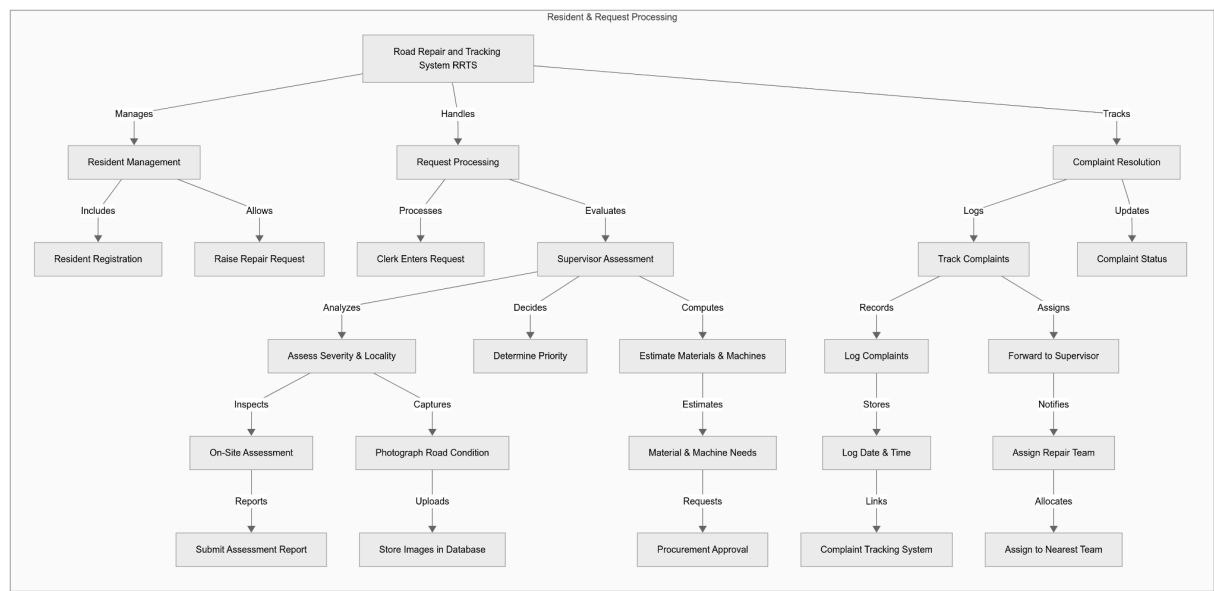


7.2 Structure Chart

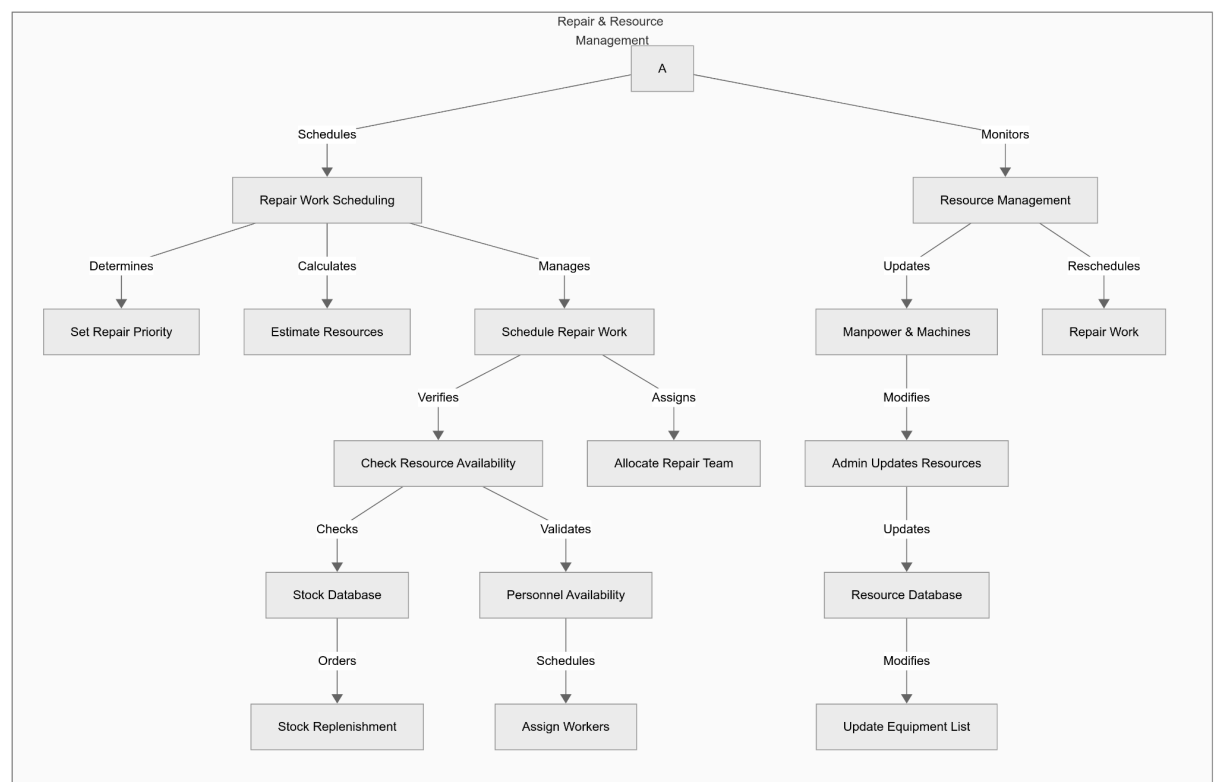
Overall Structure Chart for Road Repair and Tracking System



Structure Chart For Resident Management,Request Processing and Complaint Management



Structure Chart For Repair and Resource Management



7.3 Use Case Diagram

The Use Case Diagram for the Road Repair and Tracking System (RRTS) visually represents the interactions between different actors (users) and the system functionalities. The key actors in this system include:

1. **Resident** – The general public who report road issues.
2. **Supervisor** – Manages complaints and oversees repair work.
3. **City Admin** – Handles resource allocation and scheduling.
4. **Mayor** – Monitors road repair progress through reports.

Main Use Cases and Their Relationships

Resident's Actions:

- **Login into the RRTS Software** – Allows users to access the system.
- **Raise Complaints** – Enables residents to report road issues.

Supervisor's Responsibilities:

- **Access All Complaints** – The supervisor reviews reported issues.
- **Check Road Severity** – Extends from accessing complaints to assess the problem.
- **Assign Manpower and Machines for Road Repair** – Includes Update Repair Status to track progress.

City Admin's Actions:

- **Allot Resources** – Includes Check Resources to ensure availability.
- **Update Repair Schedule** – Extends Check Repair Schedule to manage repair timelines.

Mayor's Role:

- **Road Repair Reports** – The mayor monitors repair work progress through reports.

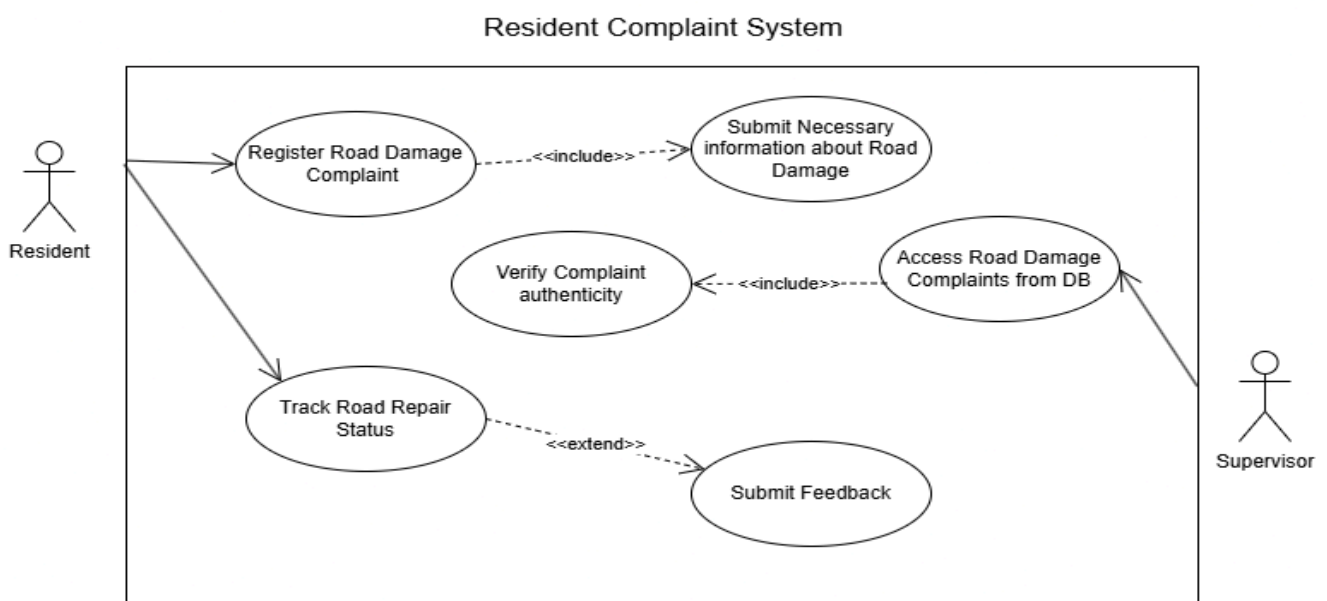
Relationships Between Use Cases

- **"Extends" Relationship:**
 - **Check Road Severity extends Access All Complaints** – This additional step occurs only if a complaint is serious.

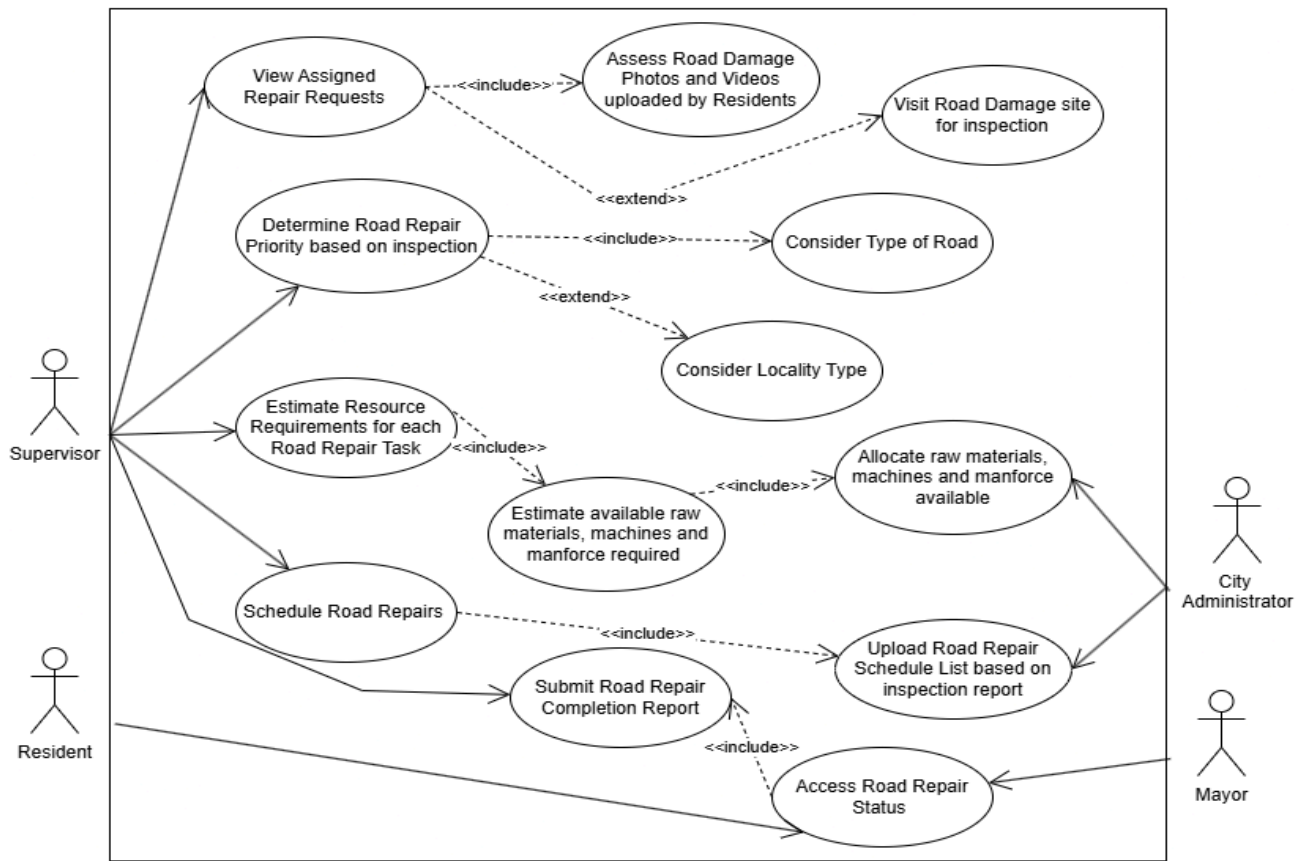
- **Check Repair Schedule extends Update Repair Schedule** – The admin verifies the schedule before updating it.
- **Check Resources extends Allot Resources** – Ensures resources are available before allotting them.
- **"Includes" Relationship:**
 - **Assign Manpower and Machines for Road Repair includes Update Repair Status** – Assigning workers inherently involves updating progress.
 - **Allot Resources includes Assign Manpower and Machines for Road Repair** – Resource allocation is needed for assigning tasks.

The **Use Case Diagram** provides a high-level overview of the system's functionalities, highlighting how different users interact with the **Road Repair and Tracking System (RRTS)**. The diagram ensures clarity in system design, aiding developers in understanding essential workflows.

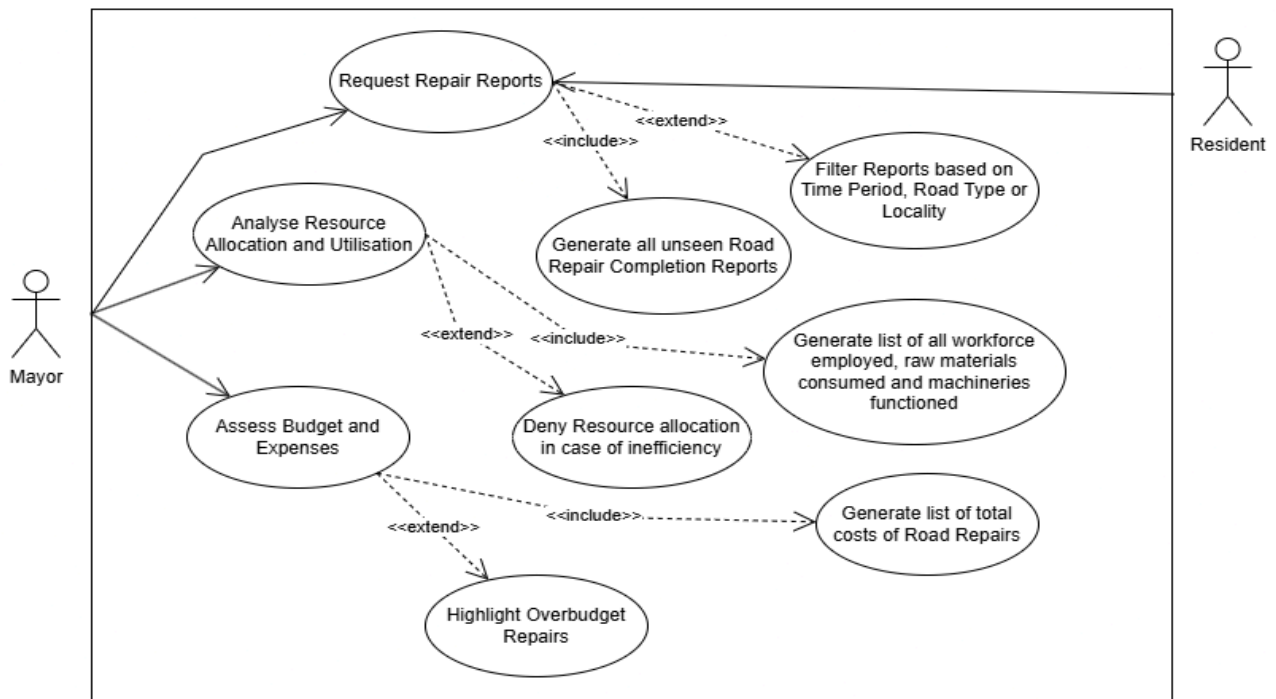
Use Case Diagrams



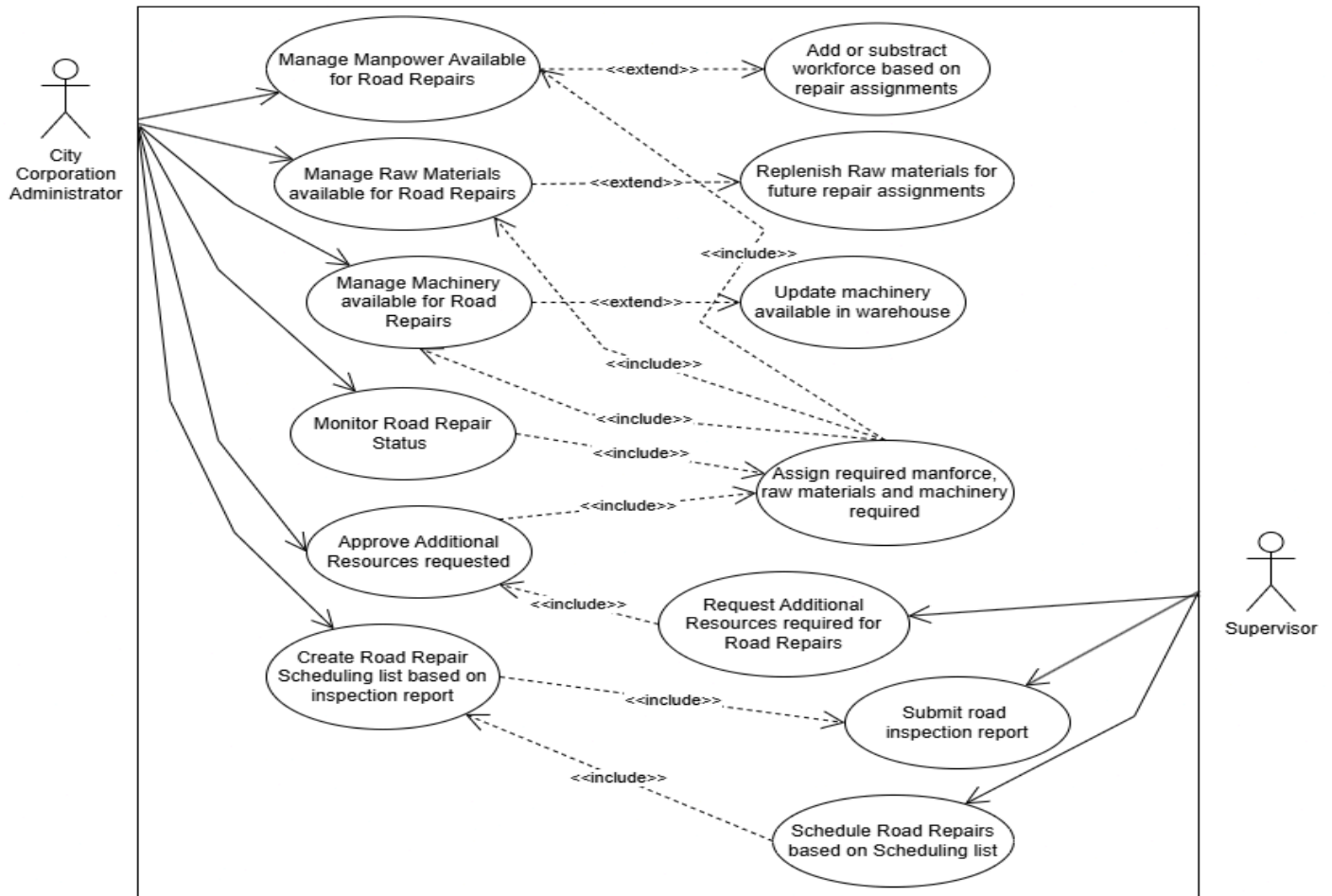
Supervisor Inspection and Road Repair Scheduling Module

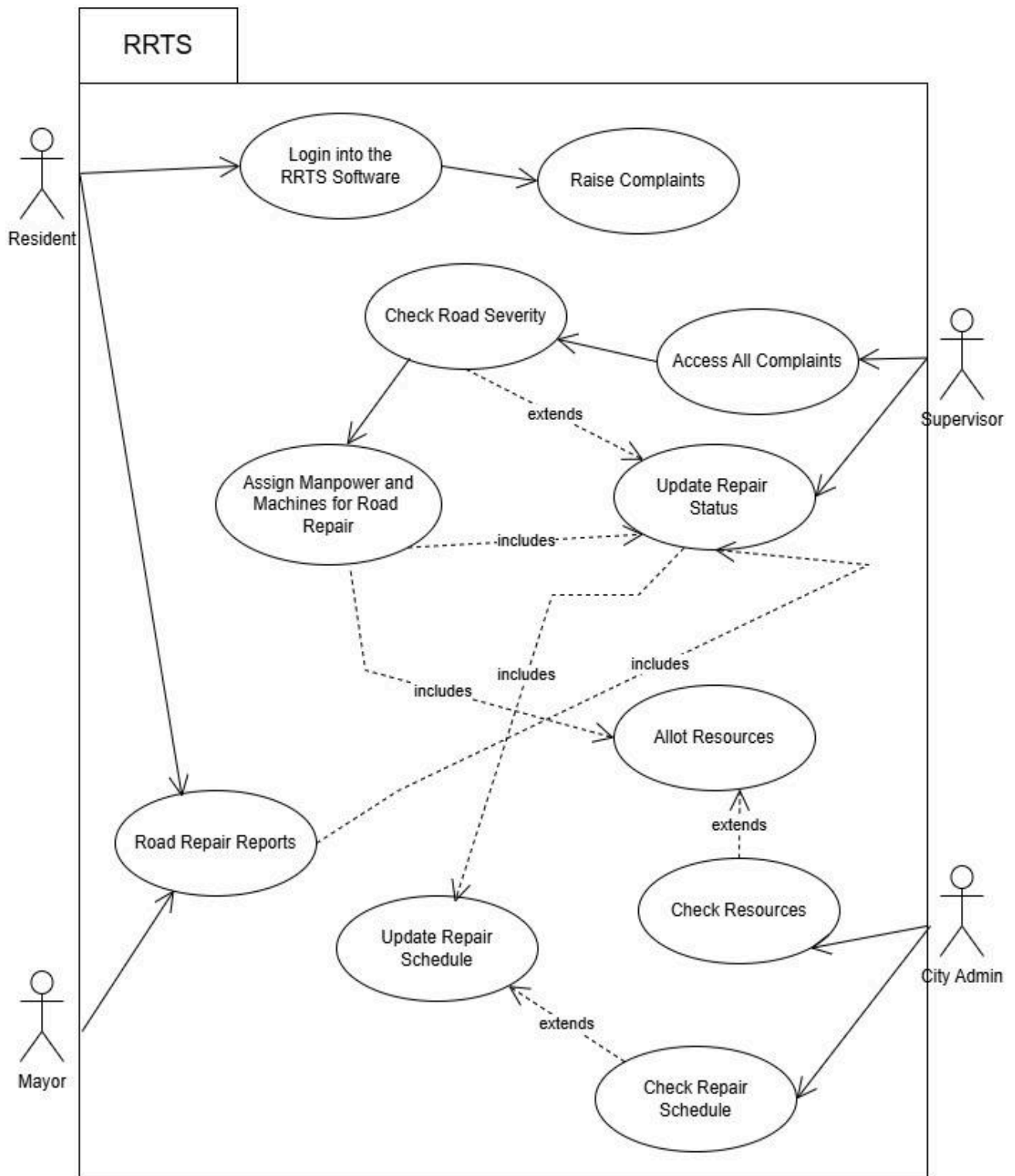


Road Repair Analysis and Tracking System

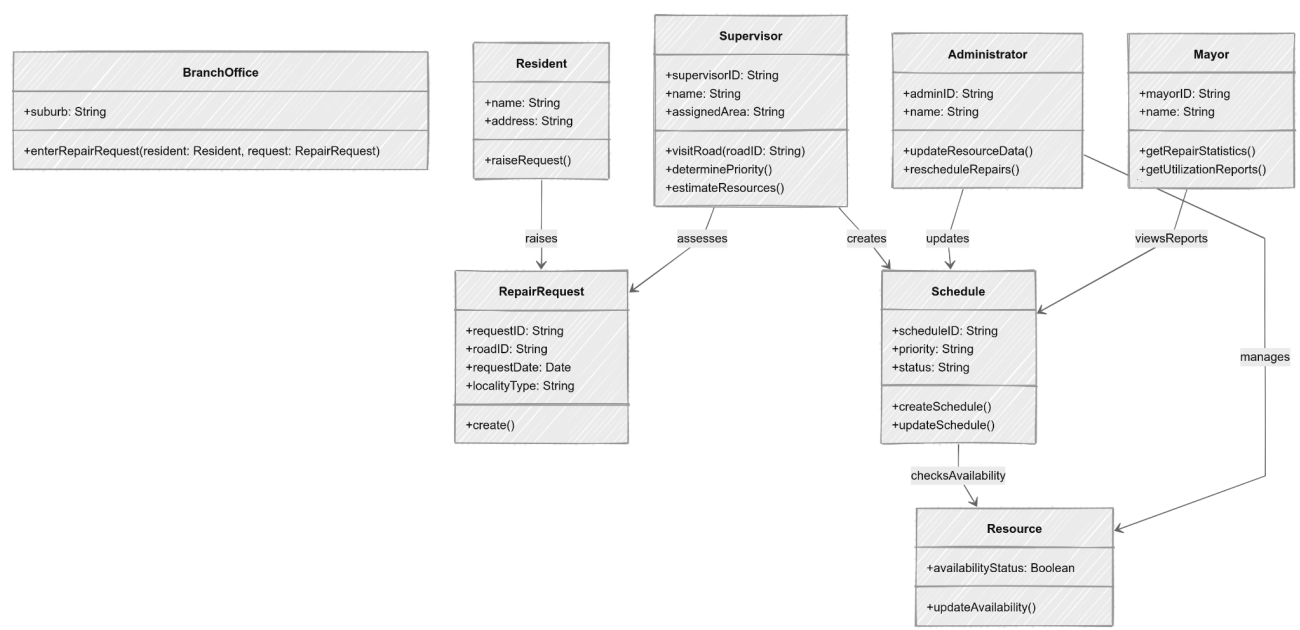


City Corporation Administrator Resource Allocation and Repair Scheduling Module

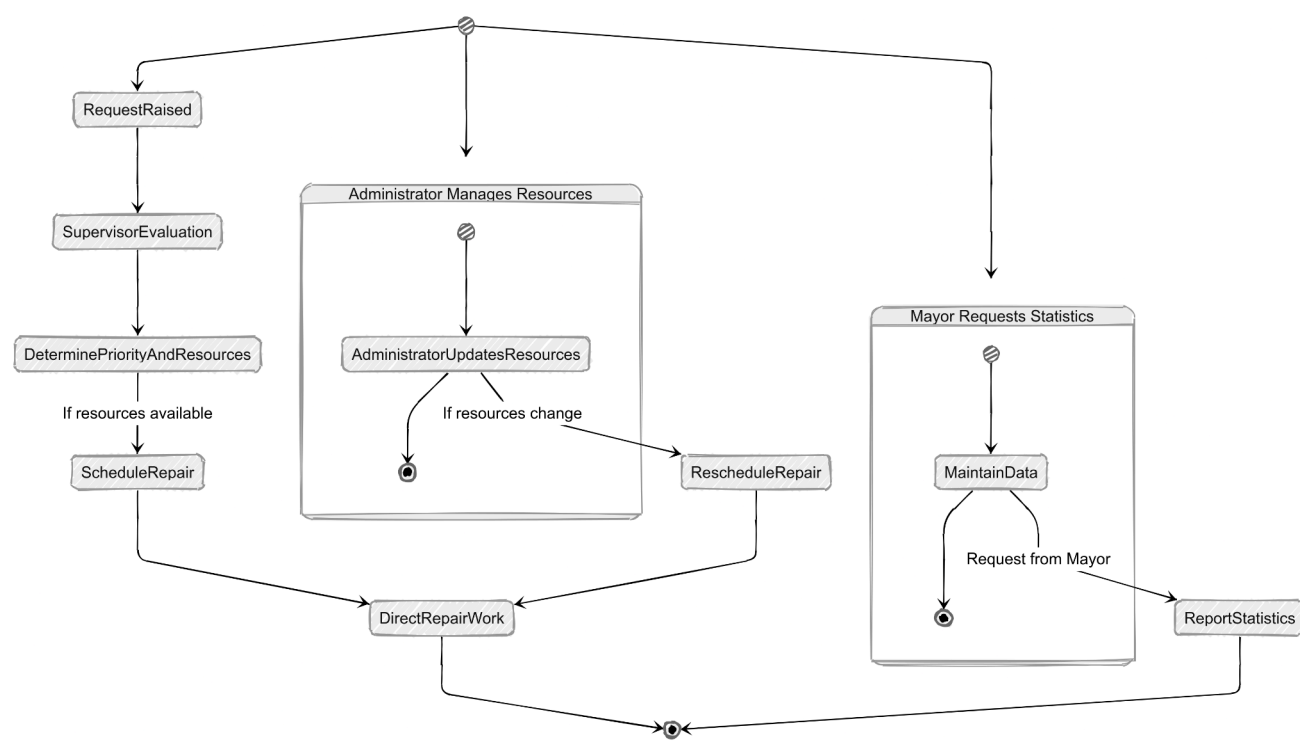




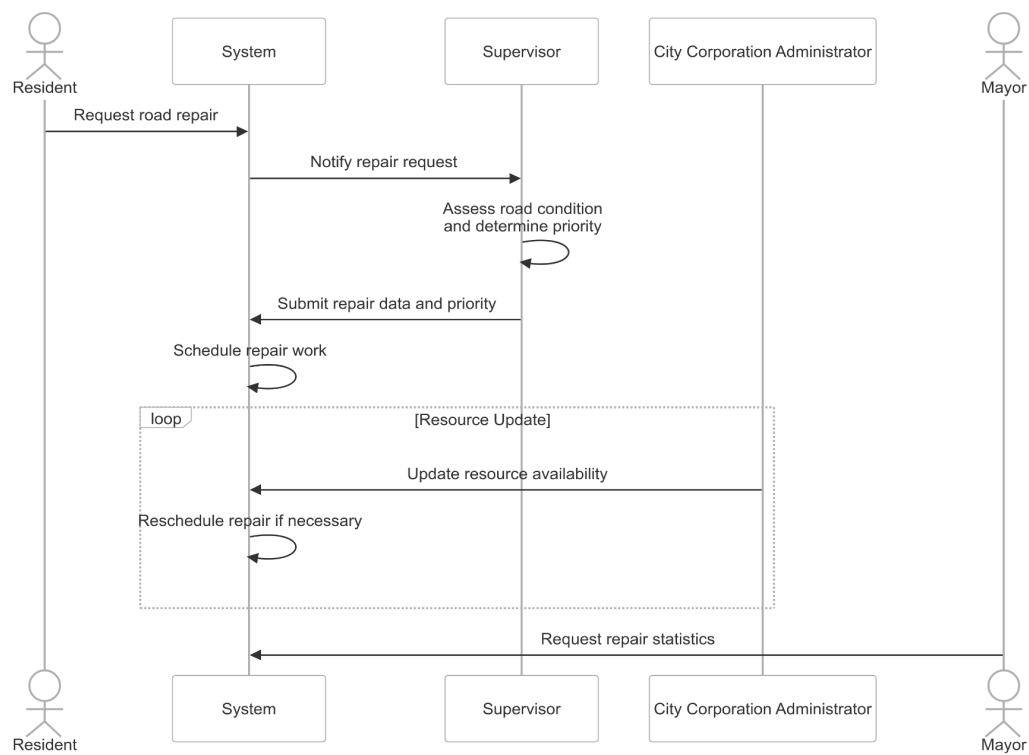
7.4 Class Diagram



7.5 Activity Diagram



7.6 Sequence Diagram



8. Conclusion

The **Road Repair and Tracking System (RRTS)** enhances public engagement by allowing citizens to raise repair requests, ensuring their concerns are addressed promptly and transparently.

The system is highly scalable and flexible, adapting to increasing repair demands and fluctuating resource availability. With data-driven decision-making, historical repair data helps city officials plan better maintenance strategies and optimize budgets.

Real-time progress tracking enhances accountability, ensuring timely completion and minimizing resource wastage. The system features a user-friendly interface, making it easy for clerks, supervisors, and administrators to adopt and use effectively.

Benefits of the Road Repair and Tracking Software (RRTS)

1. **Efficient Request Handling** – Residents can raise repair requests that are logged systematically, reducing paperwork and ensuring no request is overlooked.
2. **Automated Prioritization** – The system prioritizes road repairs based on severity, locality type, and urgency, ensuring critical roads are repaired first.
3. **Resource Optimization** – Helps in efficient allocation of raw materials, machinery, and personnel based on availability and priority.
4. **Real-Time Scheduling** – Dynamically schedules repair work based on available resources, ensuring faster response times and reducing project delays.
5. **Improved Monitoring and Tracking** – Supervisors can track the progress of ongoing repairs, manage timelines, and make necessary adjustments.
6. **Data-Driven Decision Making** – Provides the mayor and city officials with detailed reports on completed, ongoing, and pending repairs for better governance.
7. **Transparency and Accountability** – Maintains a digital record of all repairs, ensuring accountability among workers and supervisors.
8. **Enhanced Communication** – Facilitates better coordination between clerks, supervisors, and administrators, reducing miscommunication and delays.
9. **Historical Data Analysis** – Allows the government to analyze past road conditions, identify patterns, and plan preventive maintenance strategies.

